

WEEK-3 - BOOKFLOW

```
test> use BookFlow
switched to db BookFlow
BookFlow> db.createCollection("book_metadata")
{ ok: 1 }
BookFlow> show collections
book_metadata
BookFlow> db.book_metadata.insertOne({
  title: "The Silent Algorithm",
  author: "R. Menon",
  genre: "Technology",
  publishedYear: 2022,
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: "K. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
})
{
  acknowledged: true,
  insertedId: ObjectId('6aad4ea63ee6308fe61e596d')
}
BookFlow> db.book_metadata.insertMany([
```

```
mongosh mongodb://127.0.0.1:27027/BookFlow> db.book_metadata.insertMany([
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: ISODate("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: ISODate("2022-06-01")
      }
    ]
  },
  {
    title: "Whispers of Yesterday",
    author: "S. Kapoor",
    genre: "Fiction",
    publishedYear: 2019,
    formats: ["Hardcover", "Paperback"],
    specs: {
      pages: 410,
      binding: "Sewn"
    },
    avgRating: 3.8,
    reviews: [
      {
```

```
mongosh mongodb://127.0.0.1:27027/ > use books
books> insertMany([
  {
    member_id: "M3001",
    rating: 3,
    comment: "Slow start but strong ending.",
    date: ISODate("2020-01-15")
  },
  {
    member_id: "M3002",
    rating: 4,
    comment: "Beautifully written prose.",
    date: ISODate("2020-02-11")
  }
],
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: {
    fileSize_MB: 5.6,
    drm: true,
    narrator: "P. Iyer",
    runtime_min: 540
  },
  avgRating: 4.9,
  reviews: [
    {
      member_id: "M4001",
      rating: 5,
      comment: "The Digital format made annotations so easy.",
      date: ISODate("2023-08-01")
    },
    {
      member_id: "M4002",
      rating: 5,
      comment: "Mind-bending and well narrated.",
      date: ISODate("2023-09-14")
    }
  ]
}]
```

```
mongosh mongodb://127.0.0.1:27021/BookFlow?authSource=admin
> use BookFlow
> db.book_metadata.insertMany([
  {
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: ['Hardcover', 'Digital', 'Audiobook'],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',

```

```
mongosh mongodb://127.0.0.0. x + v
comment: 'Good but the Digital edition had formatting issues.',
date: ISODate('2023-02-20T00:00:00.000Z')
}
]
},
{
  _id: ObjectId('6aad4eee3ee6308fe61e596e'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6aad4eee3ee6308fe61e596f'),
  title: 'Whispers of Yesterday',
  author: 'S. Kapoor',
  genre: 'Fiction',
  publishedYear: 2019,
  formats: [ 'Hardcover', 'Paperback' ],
  specs: { pages: 410, binding: 'Sewn' },
  avgRating: 3.8,
  reviews: [
    {
      member_id: 'M3001',
```

```
mongosh mongodb://127.0.0.1 × + ▾  
  
    rating: 3,  
    comment: 'Slow start but strong ending.',  
    date: ISODate('2020-01-15T00:00:00.000Z')  
  },  
  {  
    member_id: 'M3002',  
    rating: 4,  
    comment: 'Beautifully written prose.',  
    date: ISODate('2020-02-11T00:00:00.000Z')  
  }  
]  
},  
{  
  _id: ObjectId('6aad4eee3ee6308fe61e5970'),  
  title: 'Quantum Horizons',  
  author: 'R. Menon',  
  genre: 'Science',  
  publishedYear: 2023,  
  formats: [ 'Digital', 'Audiobook' ],  
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min:  
40 },  
  avgRating: 4.9,  
  reviews: [  
    {  
      member_id: 'M4001',  
      rating: 5,  
      comment: 'The Digital format made annotations so easy.',  
      date: ISODate('2023-08-01T00:00:00.000Z')  
    },  
    {  
      member_id: 'M4002',  
      rating: 5,  
      comment: 'Mind-bending and well narrated.',  
      date: ISODate('2023-09-14T00:00:00.000Z')  
    }  
  ]  
}  
  
bookFlow> db.book_metadata.find()  
{
```

```
mongosh mongodb://127.0.0.1:27027/BookFlow> db.book_metadata.find()
[
  {
    _id: ObjectId('6aad4ea63ee6308fe61e596d'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aad4eee3ee6308fe61e596e'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
  }
]
```

```
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aad4eee3ee6308fe61e596f'),
    title: 'Whispers of Yesterday',
    author: 'S. Kapoor',
    genre: 'Fiction',
    publishedYear: 2019,
    formats: [ 'Hardcover', 'Paperback' ],
    specs: { pages: 410, binding: 'Sewn' },
    avgRating: 3.8,
    reviews: [
      {
        member_id: 'M3001',
        rating: 3,
        comment: 'Slow start but strong ending.',
        date: ISODate('2020-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M3002',
        rating: 4,
        comment: 'Beautifully written prose.',
        date: ISODate('2020-02-11T00:00:00.000Z')
      }
    ]
  }
],
}
```

```
{
  _id: ObjectId('6aad4eee3ee6308fe61e5970'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}

BookFlow> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)

{ title: 'The Silent Algorithm', avgRating: 4.6 },
{ title: 'Ocean of Code', avgRating: 4.2 },
{ title: 'Quantum Horizons', avgRating: 4.9 }

BookFlow> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
)

{
  title: 'The Silent Algorithm',
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
},
{ title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
{ title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }

BookFlow> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
```

```
mongosh mongodb://127.0.0.1:27027/ > use books
switched to db books
books> find({})
{
  "_id": ObjectId("6a4d4ea63ee6308fe61e596d"),
  "title": "The Silent Algorithm",
  "author": "R. Menon",
  "genre": "Technology",
  "publishedYear": 2022,
  "formats": [ "Hardcover", "Digital", "Audiobook" ],
  "specs": {
    "pages": 340,
    "fileSize_MB": 4.2,
    "drm": true,
    "narrator": "K. Rao",
    "runtime_min": 610
  },
  "avgRating": 4.6,
  "reviews": [
    {
      "member_id": "M1001",
      "rating": 5,
      "comment": "Excellent deep dive into digital systems design.",
      "date": ISODate("2023-01-15T00:00:00.000Z")
    },
    {
      "member_id": "M1042",
      "rating": 4,
      "comment": "Good but the Digital edition had formatting issues.",
      "date": ISODate("2023-02-20T00:00:00.000Z")
    }
  ]
},
{
  "_id": ObjectId("6a4d4eee3ee6308fe61e596e"),
  "title": "Ocean of Code",
  "author": "A. Fernandes",
  "genre": "Technology",
  "publishedYear": 2021,
  "formats": [ "Digital", "Paperback" ],
  "specs": { "pages": 210, "fileSize_MB": 2.1, "drm": false },
  "avgRating": 4.2,
  "reviews": [
    {
      "member_id": "M2001",
      "rating": 4,
      "comment": "Solid Digital read, clean layout.",
      "date": ISODate("2022-05-10T00:00:00.000Z")
    }
  ]
}
```

```
{
  member_id: 'M2002',
  rating: 5,
  comment: 'Best programming book this year.',
  date: ISODate('2022-06-01T00:00:00.000Z')
}
]
},
{
  _id: ObjectId('6aad4eee3ee6308fe61e5970'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]
BookFlow> db.book_metadata.find(
|   { formats: { $regex: "Digital", $options: "i" } },
|   { title: 1, formats: 1, _id: 0 }
| )
|
| [
|   {
|     title: 'The Silent Algorithm',
|     formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
|   },
|   { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
|   { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
| ]
BookFlow> db.book_metadata.find(
|   { "reviews.comment": { $regex: "Digital", $options: "i" } },
|   { title: 1, "reviews.$": 1, _id: 0 }
| )
|
```

```

    rating: 5,
    comment: 'Excellent deep dive into digital systems design.',
    date: ISODate('2023-01-15T00:00:00.000Z')
  }
],
{
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    }
  ]
},
{
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    }
  ]
}
]
BookFlow> db.book_metadata.aggregate([
|   { $match: { publishedYear: { $gt: 2020 } } }
| ])
| [
|   {
|     _id: ObjectId('6aad4ea63ee6308fe61e596d'),
|     title: 'The Silent Algorithm',
|     author: 'R. Menon',
|     genre: 'Technology',
|     publishedYear: 2022,
|     formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|     specs: {
|       pages: 340,
|       fileSize_MB: 4.2,
|       drm: true,
|       narrator: 'K. Rao',
|       runtime_min: 610
|     },
|     avgRating: 4.6,
|     reviews: [

```

```
{
  member_id: 'M1001',
  rating: 5,
  comment: 'Excellent deep dive into digital systems design.',
  date: ISODate('2023-01-15T00:00:00.000Z')
},
{
  member_id: 'M1042',
  rating: 4,
  comment: 'Good but the Digital edition had formatting issues.',
  date: ISODate('2023-02-20T00:00:00.000Z')
}
]
},
{
  _id: ObjectId('6aad4eee3ee6308fe61e596e'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ]
},
{
  _id: ObjectId('6aad4eee3ee6308fe61e5970'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
```

```
{
  member_id: 'M4001',
  rating: 5,
  comment: 'The Digital format made annotations so easy.',
  date: ISODate('2023-08-01T00:00:00.000Z')
},
{
  member_id: 'M4002',
  rating: 5,
  comment: 'Mind-bending and well narrated.',
  date: ISODate('2023-09-14T00:00:00.000Z')
}
]
}
]
BookFlow> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" }
])
{
  _id: ObjectId('6aad4ea63ee6308fe61e596d'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: {
    member_id: 'M1001',
    rating: 5,
    comment: 'Excellent deep dive into digital systems design.',
    date: ISODate('2023-01-15T00:00:00.000Z')
  }
},
{
  _id: ObjectId('6aad4ea63ee6308fe61e596d'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
```

```
mongosh mongodb://127.0.0.1:27021/
> use books
switched to db books
> find({})
{
  "_id": ObjectId("6a4d4eee3ee6308fe61e596e"),
  "title": "Ocean of Code",
  "author": "A. Fernandes",
  "genre": "Technology",
  "publishedYear": 2021,
  "formats": [ "Digital", "Paperback" ],
  "specs": { "pages": 210, "fileSize_MB": 2.1, "drm": false },
  "avgRating": 4.2,
  "reviews": {
    "member_id": "M2001",
    "rating": 4,
    "comment": "Solid Digital read, clean layout.",
    "date": ISODate("2022-05-10T00:00:00.000Z")
  }
},
{
  "_id": ObjectId("6a4d4eee3ee6308fe61e596e"),
  "title": "Ocean of Code",
  "author": "A. Fernandes",
  "genre": "Technology",
  "publishedYear": 2021,
  "formats": [ "Digital", "Paperback" ],
  "specs": { "pages": 210, "fileSize_MB": 2.1, "drm": false },
  "avgRating": 4.2,
  "reviews": {
    "member_id": "M2002",
    "rating": 5,
    "comment": "Best programming book this year.",
    "date": ISODate("2022-06-01T00:00:00.000Z")
  }
},
{
  "_id": ObjectId("6a4d4eee3ee6308fe61e5970"),
  "title": "Quantum Horizons",
  "author": "R. Menon",
  "genre": "Science",
  "publishedYear": 2023,
  "formats": [ "Digital", "Audiobook" ],
  "specs": { "fileSize_MB": 5.6, "drm": true, "narrator": "P. Iyer", "runtime_min": 540 },
  "avgRating": 4.9,
  "reviews": {
    "member_id": "M4001",
    "rating": 5,
    "comment": "The Digital format made annotations so easy.",
    "date": ISODate("2023-08-01T00:00:00.000Z")
  }
},
{
  "_id": ObjectId("6a4d4eee3ee6308fe61e5970"),
  "title": "Quantum Horizons",
  "author": "R. Menon",
  "genre": "Science",
  "publishedYear": 2023,
  "formats": [ "Digital", "Audiobook" ],
  "specs": { "fileSize_MB": 5.6, "drm": true, "narrator": "P. Iyer", "runtime_min": 540 },
  "avgRating": 4.9,
  "reviews": {
    "member_id": "M4002",
    "rating": 4,
    "comment": "Audiobook is great, but digital format is better.",
    "date": ISODate("2023-09-01T00:00:00.000Z")
  }
}
```

```

mongosh mongodb://127.0.0.1
[
  {
    $unwind: "$reviews",
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  }
]
[
  {
    _id: 'Ocean of Code',
    avgReviewRating: 4.5,
    totalReviews: 2,
    author: 'A. Fernandes'
  },
  {
    _id: 'The Silent Algorithm',
    avgReviewRating: 4.5,
    totalReviews: 2,
    author: 'R. Menon'
  },
  {
    _id: 'Quantum Horizons',
    avgReviewRating: 5,
    totalReviews: 2,
    author: 'R. Menon'
  }
]
BookFlow> { $sort: { avgReviewRating: -1 } }
-1
BookFlow> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews",
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  { $sort: { avgReviewRating: -1 } },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
])
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'Quantum Horizons',
    avgReviewRating: 5
  },
  {
    totalReviews: 2,

```

mongosh mongod://127.0.0.1

```
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
},
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
},
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
]
BookFlow> db.book_metadata.createIndex({ "reviews.comment": "text" })
reviews.comment_text
BookFlow> db.book_metadata.find({
  |   $text: { $search: "Digital" }
  | })
[
  {
    _id: ObjectId('6aad4ea63ee6308fe61e596d'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6aad4eee3ee6308fe61e5970'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  }
]
```

```
mongosh mongodb://127.0.0.0 x + v
avgRating: 4.9,
reviews: [
  {
    member_id: 'M4001',
    rating: 5,
    comment: 'The Digital format made annotations so easy.',
    date: ISODate('2023-08-01T00:00:00.000Z')
  },
  {
    member_id: 'M4002',
    rating: 5,
    comment: 'Mind-bending and well narrated.',
    date: ISODate('2023-09-14T00:00:00.000Z')
  }
]
},
{
  _id: ObjectId('6aad4eee3ee6308fe61e596e'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [ 'Digital', 'Paperback' ],
  specs: { pages: 210, fileSize_MB: 2.1, drm: false },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: ISODate('2022-05-10T00:00:00.000Z')
    },
    {
      member_id: 'M2002',
      rating: 5,
      comment: 'Best programming book this year.',
      date: ISODate('2022-06-01T00:00:00.000Z')
    }
  ]
}
]
}
BookFlow> db.book_metadata.find(
|   $text: { $search: "Digital" }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    }
  },
  indexFilterSet: false,
  queryHash: '4FB85230',
  planCacheShapeHash: '4FB85230',
  planCacheKey: 'F34F3876',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
```

mongosh mongod://127.0.0.1

```
maxIndexedOrSolutionsReached: false,
maxIndexedAndSolutionsReached: false,
maxScansToExplodeReached: false,
prunedSimilarIndexes: false,
winningPlan: {
  isCached: false,
  stage: 'TEXT_MATCH',
  nss: 'BookFlow.book_metadata',
  indexPrefix: {},
  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [ 'digit' ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  inputStage: {
    stage: 'FETCH',
    nss: 'BookFlow.book_metadata',
    inputStage: {
      stage: 'IXSCAN',
      nss: 'BookFlow.book_metadata',
      keyPattern: { _fts: 'text', _ftsx: 1 },
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
      indexBounds: {}
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 5,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [ 'digit' ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    }
  }
}
```

```

mongosh mongodb://127.0.0.1
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {},
    keysExamined: 3,
    seeks: 1,
    dupsTested: 3,
    dupsDropped: 0,
    peakTrackedMemBytes: 26
  }
}
},
queryShapeHash: '69E5F591B2A80418301CE1729E6C89F575F3AC5A7AB1AFB697780349FD8C89E5',
peakTrackedMemBytes: Long('26'),
command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'Digital' } },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'Samhita',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
BookFlow> db.book_metadata.find({
|   $text: { $search: "Digital" }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    }
  },
  indexFilterSet: false,
  queryHash: '4FB85230',
  planCacheShapeHash: '4FB85230',
  planCacheKey: 'F34F3876',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,

```

```
isCached: false,
stage: 'TEXT_MATCH',
nss: 'BookFlow.book_metadata',
indexPrefix: {},
indexName: 'reviews.comment_text',
parsedTextQuery: {
  terms: [ 'digit' ],
  negatedTerms: [],
  phrases: [],
  negatedPhrases: []
},
textIndexVersion: 3,
inputStage: {
  stage: 'FETCH',
  nss: 'BookFlow.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'BookFlow.book_metadata',
    keyPattern: { _fts: 'text', _ftsx: 1 },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  }
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 0,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'BookFlow.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [ 'digit' ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    },
    textIndexVersion: 3,
    docsRejected: 0,
    inputStage: {
      stage: 'FETCH',
```